

B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education (CBCGS-HME 2023)
Proposed TCET Autonomy Scheme (w.e.f. A.Y. 2024-25)

B.E. Information Technology					B.E. (SEM : VII)					
Course Name: Machine Learning for IT Application Development					Course Code: PCC- IT 701					
Teaching Scheme (Program Specific)					Examination Scheme (Academic)					
Modes of Teaching / Learning / Weightage					Modes of Continuous Assessment / Evaluation					
Hours Per Week					Theory (100)		Practical/ Oral (25)	Term Work (25)	Total	
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR	TW	
3	-	2	5	4	20	20	60	25	25	150
ISE: In-Semester Examination - Paper Duration – 1 Hour ESE: End Semester Examination - Paper Duration - 2 Hours Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).										
Prerequisite: Linear Algebra, Calculus, Probability, Statistics.										

Course Objective:

The course intends to deliver the fundamentals of Machine Learning, in particular focusing on the core concepts of supervised, unsupervised, semi supervised and reinforcement learning. Students will learn the algorithms which underpin many popular Machine Learning techniques, as well as developing an understanding of the hypothetical relationships between these algorithms.

Course Outcomes: Upon completion of the course students will be able to:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy	PO Mapping	PSO Mapping
1	Understand basics of Machine Learning	L1, L2, L3	2,3	1,2
2	Apply pre-processing techniques	L1, L2, L3	1,2,4	1,2
3	Understand and apply regression for learning and assess the outcome	L1, L2, L3, L4	1,2,3,4	1,2
4	Apply classification for learning and assess the outcome	L1, L2, L3, L4	1,2,3,4	1,2
5	Apply optimization techniques for performance enhancement	L1, L2, L3, L4	2,4,5	1,2
6	Illustrate semi supervised and reinforcement learning concepts	L1, L2, L3, L4	2,4	1,2

Detailed Syllabus:

Module No.	Topics	Hrs	Cognitive levels of attainment as per Bloom's Taxonomy
1	Introduction to Machine Learning	5	L1, L2
	Machine Learning terminology, Types of Machine Learning, Issues in Machine Learning, Application of Machine Learning, Steps in developing ML application, how to choose the right algorithm, Hypothesis Testing		
2	Data Handling Techniques	6	L1, L2, L3
	Data Handling Techniques, Introduction to Feature selection and extraction, Feature Selection and Extraction Methods, Imbalanced Dataset Handling Techniques		
3	Regression Techniques and Advanced Clustering	10	L1, L2, L3, L4
	Simple and multiple Linear Regression, Logistic Regression, Regularization, Evaluation Metric, Probabilistic Model-Based Clustering, BIRCH, Use cases		
4	Classification Techniques	10	L1, L2, L3, L4
	k Nearest Neighbour, Support Vector Machine, Decision Tree (CART), Issues in Decision Tree, Difference between Regression and classification, Ensembles Techniques -Bagging, Boosting, Evaluation Metrics, Use cases		
5	Optimization Techniques	7	L1, L2, L3, L4
	Model Selection techniques, Cross Validation, Grid Search method, Gradient descent, Types of Gradient descent, Hyper parameter tuning		
6	Semi Supervised Learning & Reinforcement Learning	7	L1, L2, L3, L4
	Semi-supervised Learning, Benefits and limitations, Semi-supervised Learning Techniques Reinforcement Learning: Basics, Elements, Algorithms, Use cases		
Total Hours		45	

Books and References:

Sr. No	Title	Authors	Publisher	Edition	Year
1	Machine Learning In Action	Peter Harrington	DreamTech Press	1 st	2012
2	Introduction to Machine Learning	Ethem Alpaydın	MIT Press	4 th	2020

3	Machine Learning	Tom M. Mitchell	McGraw Hill	2 nd	1997
4	Machine Learning An Algorithmic Perspective	Stephen Marsland	CRC Press	2 nd	2011
5	Machine Learning — A Probabilistic Perspective	Kevin P. Murphy	MIT Press	1 st	2012
6	Pattern Recognition and Machine Learning	Christopher M. Bishop	Springer	-	2006
7	Elements of Statistical Learning	Trevor Hastie, Robert Tibshirani, Jerome Friedman	Springer	-	2017

Online References:

Sr. No.	Website Name	URL	Modules Covered
1	www.analyticvidhya.com	https://www.analyticsvidhya.com/%20machine%20learning/	M1-M6
2	www.towardsdatascience.com	https://towardsdatascience.com/machine-learning/home	M1-M6
3	www.coursera.org	https://www.coursera.org/learn/machine-learning	M1-M6